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SUBJECT CODE NO: - RNP-8001

FACULTY OF SCIENCE AND TECHNOLOGY

F.E. (Group A & B) (NEP) Sem - I

Examination November / December 2025 (To be held in February-2026)

Applied Mathematics-I

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions:

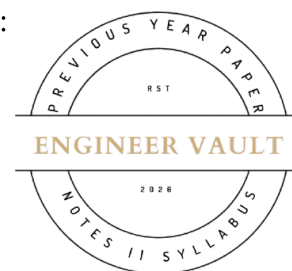
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- 1) A circuit containing resistance R and inductance L in series with voltage source E then the D.E for current is:
 - a) $Li + R \frac{di}{dt} = E$
 - b) $L \frac{di}{dt} + Ri = E$
 - c) $Li - R \frac{di}{dt} = E$
 - d) $Li + R \frac{di}{dt} = 0$
- 2) The necessary & sufficient condition for D.E. to be exact D.E. is:
 - a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
 - b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$
 - c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$
 - d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$
- 3) The solution of D.E. $xy(1 + xy^2) \frac{dy}{dx} = 1$ is:
 - a) $x + y^2 - 2 = ce^x$
 - b) $\frac{1}{x} + y^2 - 2 = ce^{-\frac{y^2}{2}}$
 - c) $xy^2 - 2 = ce^x$
 - d) $\frac{1}{x} + 2y^2 = ce^{-\frac{y^2}{2}}$
- 4) The rate of change of velocity is given by:
 - a) $\frac{dv}{dt}$
 - b) $\frac{dv}{dx}$
 - c) $a \frac{dv}{dx}$
 - d) $m \frac{dv}{dx}$
- 5) If $x = e^\theta \cos \phi$ $y = e^\theta \sin \phi$ then $\frac{\partial(x,y)}{\partial(\theta,\phi)}$ is:
 - a) $x + y$
 - b) $x^2 + y^2$
 - c) e^θ
 - d) 1
- 6) The stationary points of the function $f(x, y) = x^3 + y^3 - 3axy$ are:
 - a) $x = 0, y = a$
 - b) $x = a, y = 0$
 - c) $x = a, y = a$
 - d) $x = a, y = -a$

- 7) If z is homogeneous function of x, y of degree n & $z = f(u)$ then by Euler's thm $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial y^2} + y^2 \frac{\partial^2 u}{\partial y^2}$ is equal to:
 a) 0 b) $g(u)[g'(u) - 1]$ c) nu d) u
- 8) If $z = ax^2 + 2by^2 + 2bxy$ then $\frac{\partial z}{\partial x}$:
 a) $3by$ b) $2ax$ c) $3(ax + by)$ d) $2(ax + by)$
- 9) The value of $\lim_{x \rightarrow y} \frac{x^y - y^x}{x^x - y^y}$ is:
 a) 0 b) 1 c) ∞ d) $\frac{1 - \log y}{1 + \log y}$
- 10) By ratio test the series $\frac{2x}{1^2} + \frac{3^2 x^2}{2^3} + \frac{4^3 x^3}{3^3} + \frac{5^4 x^4}{4^3} + \dots$ _____ converges or diverges:
 a) Convergent $\frac{1}{x} > 1$ b) divergent $\frac{1}{x} < 1$
 c) Test fails $\frac{1}{x} = 1$ d) convergent $\frac{1}{x} < 0$
- 11) The expansion of series $\tan^{-1} x$ is:
 a) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ b) $1 + x + x^2 + \frac{2}{3}x^4 + \dots$
 c) $x + x^2 + \frac{2}{3}x^3 + \dots$ d) $\frac{x}{2} - \frac{x^3}{8} - \frac{1}{192}x^5 \dots \dots \dots$
- 12) Find the Taylor series expansion of the function $\cosh(x)$ at $x = 0$:
 a) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$ b) $\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty$
 c) $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \infty$ d) $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$

SECTION B

- Q2** a) Solve : $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$ 05
- b) Solve : $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$ 05
- Q3** a) If $xu = yz, yv = zx, zw = xy$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ 05
- b) If $u = \log(\tan x + \tan y + \tan z)$ then prove that : 05
- $$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$$
- Q4** a) Expand $\tan^{-1} x$ in powers of x 05
- b) Test the convergence of series $\sum \frac{n^2(n+1)^2}{n!}$ 05
- Q5** a) If $u = \sin^{-1} \sqrt{x^2 + y^2}$ then find $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$ 05



- b) Divide 24 into three parts such that the continued product of the first, square of the Second & cube of the third is maximum **05**

Q6 a) Expand $\log x$ in powers of $(x - 4)$ **05**

- b) Evaluate : $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{1}{\log(x-1)} \right]$ **05**

Q7 a) A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohms in series and a constant inductance L henries. If the initial current is zero show that the current builds up to half of its theoretical maximum in $\frac{(L \log 2)}{R}$ seconds **05**

- b) A body of mass m , falls from rest under gravity in a fluid whose resistance to motion at any instant is mK times its velocity, where K is a constant. Find the terminal velocity of the body & also the time to acquire one half of its limiting speed. **05**

